



Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Product Design with Autodesk Inventor

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Conducted by

Tertiary Infotech Academy Pte Ltd

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	16 July 2026	Legacy trainer deck and existing delivery materials.	Tertiary Infotech Academy
9.0	20 August 2026	Rebuilt visual courseware, detailed Learner Guide, eight progressive labs, native charts and aligned assessments.	Dr. Alfred Ang

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How to Use This Guide

This guide supports Product Design with Autodesk Inventor (TGS-2021006715). The slide deck explains concepts, decisions and workflows; this guide and the individual lab folders contain the detailed procedures. Complete labs in order because later evidence reuses the design-intent, assembly and documentation habits established earlier.

What You Need

- A Windows computer meeting the Autodesk Inventor requirements.
- Autodesk Inventor 2024 or later, with access to Part, Assembly, Weldment, Drawing and Presentation environments.
- The complete labs/ folder downloaded from the LMS or GitHub repository.
- Permission to save native Inventor files and screenshots in the local lab folders.

Evidence Convention

- Work from copies of supplied samples and keep all dependent files inside the current lab folder.
- Name screenshots evidence-01.png, evidence-02.png and so on.
- Record the requirement, decision, result and unresolved issue in review-notes.md.
- Use the acceptance criteria as the final self-check before submission.

Learning Outcomes and Skills Mapping

Learning outcome	Knowledge	Abilities
LO1: Conceptualise and generate machine parts using Autodesk Inventor.	K1, K2	A1, A2
LO2: Generate CAD models using Autodesk Inventor.	K3, K4	A3, A8
LO3: Produce assemblies using Autodesk Inventor.	K5	A4, A5
LO4: Connect and prepare assemblies for welding.	K6	A6, A7
LO5: Produce detailed drawings for machine parts.	K7	A1, A9, A10
LO6: Review final designs for automation and improvement.	K8	A1, A11

Topic 1 — Generate Machine Parts

Design intent · projects and files · constrained sketches · standard part features

Key concepts

Design begins with function: Translate load, motion, envelope, interface, safety and service needs into measurable acceptance criteria.

The Inventor file ecosystem: IPT parts feed IAM assemblies; IPN presentations explain assembly; IDW/DWG drawings release manufacturing intent.

Projects resolve references: An IPJ defines workspace, libraries, templates and Content Center locations so dependent files stay discoverable.

Coordinate systems anchor intent: Origin planes, axes, points and user work features give sketches stable references that survive change.

Sketch geometry is not design intent: Lines and arcs describe shape; geometric constraints describe behaviour; dimensions describe size and position.

Fully constrained is predictable: Remove unintended degrees of freedom without over-constraining; test by dragging before leaving the sketch.

Parameters make change deliberate: Meaningful names, equations and units turn dimensions into a controllable design system.

Choose the right base feature: Extrude prismatic forms, revolve axisymmetric forms, sweep a profile along a path, and loft between sections.

Placed features carry manufacturing logic: Hole, thread, fillet, chamfer, rib and shell encode operations and functional details.

Feature order affects resilience: Build stable primary form first, functional cuts next, then finishing features such as fillets and chamfers.

Standards are a design input: Select the drawing standard, units, projection convention, threads and tolerances before detailing.

Validate before reuse: Check units, materials, sketches, feature warnings, mass properties and naming before a part enters an assembly.

Working method

1. Clarify function
2. Plan references
3. Constrain sketch
4. Build features
5. Validate part

Lab 1 — Design Brief to Parametric Base Part

Mapping	LO1 · A1, A2 · K1, K2
Time	75 minutes
Scenario	Create a controllable base component for a compact universal-joint guard and prove that the sketch captures the design brief.
Deliverable	A fully constrained IPT part, a one-page design-intent note and evidence screenshots.

Detailed step-by-step procedure

1. Read the functional brief and record the mounting envelope, interface faces, minimum wall, edge-clearance and service-access requirements.

2. Start Inventor and create or activate a single-user project whose workspace is this lab folder; confirm the project points to the local samples directory.
3. Open Base Form Start.ipt, use Save As to create Lab01-Guard-Base.ipt, and confirm millimetres are the active units.
4. Inspect the Origin folder and choose the XY plane as the primary sketch plane so the central symmetry of the part is stable.
5. Create a centre rectangle for the base footprint and constrain its centre coincident with the origin.
6. Add horizontal, vertical, equal and symmetry constraints before applying dimensions; drag the geometry and identify any remaining degrees of freedom.
7. Apply named dimensions for overall length, width and mounting-edge offset; use the Parameters dialog to rename them GuardLength, GuardWidth and EdgeOffset.
8. Finish the sketch and extrude the base using a named thickness parameter; set the operation to Join.
9. Create a second sketch for the mounting holes, project only the centre reference needed, and constrain the hole points symmetrically.
10. Use the Hole feature with the specified diameter and Through All termination; do not model holes as uncontrolled extruded cuts.
11. Add the required fillets and chamfers as finishing features after the primary form and holes.
12. Assign a provisional engineering material, then update mass properties and record the mass as baseline evidence.
13. Change GuardLength by 10 mm and update the model; confirm the holes and finishing features remain correctly located.
14. Rename sketches and features in the browser, resolve every warning, and save the file.
15. Capture the fully constrained sketch, final isometric view, browser tree and parameter table for the evidence checklist.

Acceptance criteria

- ☐ Sketch reports Fully Constrained and is centred on stable origin geometry.
- ☐ Named parameters control the main dimensions without failed features.
- ☐ The IPT opens from the lab project with no missing reference or update warning.
- ☐ Evidence shows the requirement-to-feature rationale and a successful dimension change.

Lab folder: labs/lab-01-design-brief-to-parametric-base-part/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Topic 2 — Generate CAD Models

Advanced features · work geometry · freeform and sheet metal · model interoperability

Key concepts

Analyse form before modelling: Classify the part as prismatic, rotational, path-driven, section-driven, freeform or sheet metal before choosing tools.

Work geometry controls difficult features: Offset planes, tangent planes, axes and points create references where a natural face does not exist.

Projected geometry links context: Project only the edges needed; excessive external dependencies make the feature tree fragile.

Multi-body supports top-down design: Develop interacting solids in one context, then derive or make components while retaining intent.

Patterns encode repetition: Rectangular, circular and path patterns reduce duplicated work and keep repeated geometry consistent.

Mirrors encode symmetry: Mirror features or bodies across a stable plane instead of rebuilding symmetric geometry.

Freeform solves organic transitions: T-spline forms suit ergonomic and aesthetic surfaces; validate curvature and convert deliberately.

Sheet metal is rule-driven: Thickness, bend radius, relief and K-factor govern both folded geometry and manufacturable flat patterns.

Orthographic views are a test: Front, top, side and isometric views expose missing geometry, asymmetry and unclear feature relationships.

Imported CAD needs inspection: Confirm units, body type, geometry health and associativity before using STEP, IGES, SAT or Parasolid data.

Direct edits are tactical: Move face, delete face and replace face solve local changes; parametric features remain preferable for design intent.

Model health is evidence: A clean browser, resolved references, named features and stable update behaviour are part of the deliverable.

Working method

1. Classify geometry
2. Select method
3. Build references
4. Create feature
5. Verify all views

Lab 2 — Advanced Part, Sheet Metal and Orthographic Review

Mapping	LO2 · A3, A8 · K3, K4
Time	90 minutes
Scenario	Develop two alternative CAD models for a protective cover: a lofted solid concept and a manufacturable folded-sheet option.
Deliverable	A parametric lofted part, a sheet-metal part with flat pattern, and orthographic verification views.

Detailed step-by-step procedure

1. Activate the lab project, open Loft Start.ipt and save a working copy named Lab02-Lofted-Cover.ipt.
2. Inspect the existing profiles and list the section order, plane offsets and intended symmetry before creating the loft.

3. Create or repair the required offset work plane and confirm its parent references are stable origin or named work features.
4. Project only the centre and boundary geometry required to position each new section; avoid projecting every visible edge.
5. Constrain and dimension each loft section; confirm there are no open loops or overlapping profiles.
6. Create the Loft feature, select sections in a consistent order and add a centreline or rail only when it improves control.
7. Use zebra/curvature or visual inspection to identify unwanted pinching; edit the section or rail instead of hiding the defect with a fillet.
8. Apply mirror or pattern features for repeated details and verify the feature count and direction.
9. Open Sheet Metal Start.ipt, save Lab02-Sheet-Cover.ipt and verify the sheet-metal rule, thickness, bend radius and relief settings.
10. Create the base face, add flanges and reliefs, and check bend directions against the design envelope.
11. Generate the flat pattern and inspect overlaps, bend lines, reliefs and material continuity.
12. Create an IDW drawing with front, top, right and isometric views for each concept; use equal scales and consistent orientation.
13. Compare the two concepts against mass, fabrication steps, service access and evidence clarity.
14. Update one governing dimension in each model and confirm the loft, flat pattern and drawing views update without error.
15. Save all files and capture the loft sections, flat pattern and orthographic sheet as assessment evidence.

Acceptance criteria

- ☐ Loft sections are fully constrained and the resulting body has no visible discontinuity or failed feature.
- ☐ Sheet-metal thickness and bend rule are consistent and the flat pattern is valid.
- ☐ Orthographic views agree with the 3D models and show the intended geometry clearly.
- ☐ The design comparison explains why one modelling method is preferred for production.

Lab folder: labs/lab-02-advanced-part-sheet-metal-and-orthographic-review/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Topic 3 — Assembly Modeling

Top-down and bottom-up structures · degrees of freedom · joints and constraints

Key concepts

Assembly structure communicates product logic: Parts, subassemblies and purchased items should mirror manufacturing, service and BOM expectations.

Bottom-up reuses proven parts: Place independently designed components, then define their relationships and allowable motion.

Top-down controls interfaces: Start from a layout or master geometry so interacting parts derive from shared functional references.

Every component starts with six DOF: Three translations and three rotations remain until grounding, constraints or joints remove them.

Constraints position geometry: Mate, flush, angle and insert relationships remove specific degrees of freedom between selections.

Joints describe motion: Rigid, rotational, slider, cylindrical, planar and ball joints capture intended movement directly.

Ground one stable reference: Ground the base component or use origin relationships so the mechanism has a predictable frame.

Constraint order matters: Constrain primary locating surfaces first, secondary alignment next, then motion limits and detail relationships.

Interference is a release blocker: Use collision and interference analysis to distinguish acceptable contact from overlapping solids.

Representations reduce complexity: Design views, positional representations and model states support review, drawings and performance.

Imported components need ownership rules: Decide whether external geometry remains associative, becomes a reference or is converted into a controlled part.

Assembly drawings tell the build story: Views, parts lists, balloons and notes communicate order, identity and critical relationships.

Working method

1. Place base
2. Add component
3. Remove DOF
4. Test motion
5. Document assembly

Lab 3 — Bottom-Up Universal Joint Assembly

Mapping	LO3 · A4, A5 · K5
Time	90 minutes
Scenario	Assemble the supplied universal-joint components from proven part files and retain the mechanism's intended rotation.
Deliverable	A resolved IAM assembly, a DOF/constraint record and a basic assembly drawing.

Detailed step-by-step procedure

1. Activate the lab project and copy the supplied universal-joint sample folder to a working folder before editing.
2. Open Universal Joint.iam and use Resolve Link only if Inventor reports a missing file; map references to the local working folder, never to the reference folder.
3. Inspect each placed component, its occurrence name and BOM structure; identify the base, yokes, connecting rod, pin and handle.
4. Ground the base occurrence at the assembly origin and confirm that it has no unintended motion.
5. Place any missing component from the working folder and orient it near its final position.

6. Apply the primary mate or flush relationship that locates the component against its functional mounting face.
7. Apply insert or concentric relationships to align pins and cylindrical bores while retaining intended rotation.
8. Use angle or joint limits only where the design brief defines a permitted range; avoid redundant relationships.
9. Use Degrees of Freedom or drag testing after each relationship to confirm which movement was removed.
10. Drive a suitable constraint or joint through a safe range and observe the universal-joint motion.
11. Run Interference Analysis and classify each result as intended contact, fastener engagement or unacceptable overlap.
12. Create a named design view that shows the complete mechanism clearly and save the camera orientation.
13. Create an assembly drawing with base, projected and isometric views plus a parts list and balloons.
14. Save the IAM and drawing, close Inventor, then reopen the IAM from the lab project to verify all references resolve.
15. Capture the browser relationships, interference result, driven motion and assembly drawing for evidence.

Acceptance criteria

- ☐ All references resolve locally and the assembly opens without warnings.
- ☐ The base is grounded, required components are constrained, and intended rotation remains possible.
- ☐ Interference findings are reviewed and any unacceptable overlap is corrected.
- ☐ The assembly drawing includes views, parts list and balloons that agree with the IAM BOM.

Lab folder: labs/lab-03-bottom-up-universal-joint-assembly/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Lab 4 — Top-Down Layout and Assembly Relationships

Mapping	LO3 · A4, A5 · K5
Time	75 minutes
Scenario	Control a universal-shaft variant from a master layout and compare that interface-driven approach with the bottom-up assembly.
Deliverable	A layout-controlled IAM variant with documented mate, insert and angle relationships.

Detailed step-by-step procedure

1. Open Universal Shaft Start.iam from the copied layout-assembly folder and save it as Lab04-Layout-Controlled-Shaft.iam.
2. Create a layout sketch at assembly origin that defines the shaft centreline, joint centres and operating angle.
3. Name the governing dimensions ShaftSpacing and OperatingAngle and record their initial values.
4. Project or derive only the stable layout geometry required by the component interfaces.
5. Place or create the base component in context and align it to the master layout.
6. Place the remaining supplied components and apply mate/flush relationships to locate planar interfaces.
7. Apply insert relationships for pins and bores, checking that rotation is not removed unintentionally.
8. Apply the operating-angle relationship and set safe limits where appropriate.

9. Use the browser to rename relationships by function, for example PinAxis_Insert and YokeAngle_Limit.
10. Suppress one relationship at a time and observe the returned DOF; use the test to identify redundant constraints.
11. Change ShaftSpacing and OperatingAngle within the brief, update the assembly and repair any unstable reference.
12. Create a positional representation for the nominal configuration and another for the service angle.
13. Compare top-down and bottom-up approaches in the evidence sheet: ownership, change propagation, reuse and risk.
14. Create or update the assembly drawing so both configurations can be reviewed clearly.
15. Pack all dependencies in the lab folder and reopen the IAM to confirm portability.

Acceptance criteria

- ☐ Named layout parameters control the key interfaces and update the assembly predictably.
- ☐ No redundant or conflicting relationship remains in the browser.
- ☐ Nominal and service positional representations are saved and demonstrably different.
- ☐ The comparison correctly distinguishes top-down interface control from bottom-up reuse.

Lab folder: labs/lab-04-top-down-layout-and-assembly-relationships/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Topic 4 — Weldments

Preparation · weld features · post-weld machining · material and inspection decisions

Key concepts

A weldment extends an assembly: It adds ordered preparation, weld and machining groups while retaining assembly relationships.

Conversion is deliberate: A regular assembly can become a weldment, but the conversion is one-way and should follow a saved baseline.

Preparation precedes joining: Chamfers, gaps and reliefs belong in the preparation group because they exist before the weld is applied.

Weld features carry intent: Fillet, groove and cosmetic welds communicate joint type, extent and manufacturing expectation.

Machining happens after welding: Post-weld holes, facing and cuts belong after the weld group when manufacturing sequence matters.

Material choice affects the joint: Strength, weldability, corrosion, thermal response, availability and cost must be considered together.

Standards remove ambiguity: Weld symbols, sizes, lengths, pitch, finish and process notes must follow the selected standard.

BOM structure reflects fabrication: Decide whether the weldment is treated as one fabricated item or reported through its components.

Distortion is a design risk: Heat input, restraint, sequence and access can change final geometry even when the CAD nominal is perfect.

Inspection needs access: Design for visual inspection, gauges or NDT access where the risk and specification require it.

Exploded views still need judgement: A weldment may require pre-weld and post-weld communication rather than a generic exploded assembly.

Review manufacturability early: Check joint access, fit-up, tolerance stack, fixturing and post-process operations before release.

Working method

1. Prepare faces
2. Define weld
3. Add post-work
4. Inspect access
5. Release fabrication

Lab 5 — Convert an Assembly to a Weldment

Mapping	LO4 · A6, A7 · K6
Time	70 minutes
Scenario	Prepare a small fabricated bracket assembly for welding, inspection and post-weld machining using supplied assembly components.
Deliverable	A weldment IAM showing preparation, weld and machining groups plus a fabrication-review sheet.

Detailed step-by-step procedure

1. Open the copied Universal Joint.iam only as a component source and save a separate working assembly named Lab05-Welded-Bracket.iam.
2. Remove components that are not part of the fabricated bracket and position the remaining parts with clear joint access.
3. Assign candidate physical materials and compare weldability, strength, corrosion exposure, availability and cost.
4. Save a baseline copy, then choose Environments > Convert to Weldment and select the required standard and weld-bead material.
5. Enter the Preparations group and create the required edge chamfer, gap or relief before the joint is welded.
6. Enter the Welds group and create a fillet, groove or cosmetic weld that matches the joint design.
7. Set weld length, intermittent pitch or extent as required and confirm the weld direction and participating faces.
8. Enter the Machining group and add the specified post-weld face or hole operation.
9. Inspect the browser order and explain why preparation, weld and machining features must remain in sequence.
10. Check mass properties and BOM structure, then record whether the weldment reports as one fabricated item or individual components.

11. Create a drawing view and add the applicable weld symbol, material note and inspection requirement.
12. Review access for torch, fixture, inspection and post-weld tool; mark any design change needed.
13. Create a simple presentation or snapshot sequence that distinguishes before-weld, after-weld and post-machining states.
14. Save, reopen and update the weldment to confirm no failed group or missing component reference.
15. Capture preparation, weld, machining, drawing symbol and review notes for evidence.

Acceptance criteria

- ☐ The working IAM is a weldment with correctly ordered preparation, weld and machining features.
- ☐ Material choice is justified using at least four relevant criteria.
- ☐ The drawing communicates weld type, extent and inspection intent under the selected standard.
- ☐ Fabrication access and post-weld operations have been reviewed and recorded.

Lab folder: labs/lab-05-convert-an-assembly-to-a-weldment/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Topic 5 — Deliverables

Drawing views · dimensions and GD&T · parts lists · release-ready communication

Key concepts

A drawing is a controlled communication: It translates design intent into a release package that manufacturing and inspection can interpret consistently.

Templates set the contract: Sheet size, border, title block, units, projection, styles and revision fields should be correct before views are placed.

Base view establishes orientation: Choose the most descriptive orientation, scale, hidden-line treatment and design representation.

Projected views preserve alignment: Create orthographic and isometric views from the base so projection relationships stay associative.

Sections explain internal form: Use cutting planes only where they reveal otherwise hidden geometry; hatch and labels must remain clear.

Details enlarge critical areas: A detail view focuses tolerance, fit or feature complexity without crowding the parent view.

Dimension for manufacture: Select functional datums and avoid duplicated, chained or ambiguous dimensions that hide tolerance accumulation.

GD&T controls variation: Feature control frames define characteristic, tolerance zone, material condition and datum references.

Datum schemes simulate function: Primary, secondary and tertiary datums constrain the part in the sequence used for assembly or inspection.

Parts lists and balloons link identity: Item numbers, quantities, part numbers and descriptions must match the assembly BOM.

Associativity needs checking: A green model update is not proof the drawing still communicates correctly; inspect every affected view and note.

Release uses a checklist: Verify views, scales, units, tolerances, materials, notes, revision, approvals and exported deliverables.

Working method

1. Set standard
2. Place views
3. Add dimensions
4. Apply GD&T
5. Review release

Lab 6 — Production Drawing and GD&T Package

Mapping	LO5 · A9, A10 · K7
Time	90 minutes
Scenario	Create a release-ready drawing for an engine-block component using views, dimensions, tolerances and a functional datum reference frame.
Deliverable	A multi-view IDW drawing with section/detail views, dimensions, GD&T, notes and completed release checklist.

Detailed step-by-step procedure

1. Activate the lab project and open the supplied Engine Block.ipt from the copied drawing-pack folder.
2. Review the part orientation, physical material, units and model status before creating the drawing.
3. Create a new IDW from the approved standard template and verify sheet size, projection convention, units and title block.
4. Place a base view at an appropriate scale and orientation, then create aligned front, top, side and isometric projected views.
5. Create a section view that exposes the critical internal passage or bore and label the cutting plane clearly.
6. Create a detail view for the critical fit or edge feature and set a scale that supports annotation.
7. Define a functional datum reference frame: identify the primary mounting plane, secondary locating feature and tertiary clocking feature.
8. Add general dimensions from the functional datums and remove any duplicate or non-manufacturing dimension.
9. Apply baseline or ordinate dimensions to coordinate features and use chain dimensions only where accumulation is acceptable.
10. Add size tolerances and the required feature control frame, datum identifiers and material-condition modifier where justified.
11. Add surface, hole/thread, material, finish and general-tolerance notes required by the design brief.
12. Check that every annotation is attached, legible, inside the printable area and consistent with the selected standard.
13. Change one model dimension, update the drawing and inspect every affected view, dimension and annotation for unintended movement.

14. Complete the release checklist covering title block, revision, views, scales, units, tolerances, GD&T, material, notes and approval.
15. Save the IDW and export a PDF; capture the datum scheme, feature control frame and final sheet for evidence.

Acceptance criteria

- ☐ Base, projected, section, detail and isometric views are correctly aligned and readable.
- ☐ Dimensions are non-duplicated, trace to functional datums and avoid uncontrolled tolerance chains.
- ☐ GD&T is syntactically complete and the datum scheme reflects assembly or inspection function.
- ☐ The drawing updates associatively and passes the release checklist.

Lab folder: labs/lab-06-production-drawing-and-gd-t-package/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Topic 6 — Review Final Designs

Presentation and animation · material evidence · automation · design-for-manufacture review

Key concepts

Review against requirements: A design is ready only when function, interfaces, performance, manufacture, safety and evidence meet the brief.

Scenes preserve review viewpoints: Presentation scenes can show different representations, camera positions and communication purposes.

Tweaks explain assembly: Explode direction, distance, order and trail visibility should reveal relationships without implying impossible motion.

Storyboards organise time: A storyboard sequences component movement, camera actions and snapshots into a repeatable explanation.

Animation is evidence, not decoration: Use movement to prove order, clearance or function; the static drawing must still carry complete meaning.

Materials connect geometry to performance: Assign physical materials and verify density, mass, centre of gravity, appearance and manufacturing implications.

Parameters expose design controls: Named user parameters make key dimensions visible to review and enable repeatable variants.

iLogic captures rules: Rules can validate ranges, switch features, update properties and automate drawings when the intent is explicit.

The API extends automation: Inventor's object model supports documents, components, features, properties and batch workflows; govern scripts as production tools.

DFMA reduces lifecycle cost: Simplify part count, handling, fastening, tool access and process steps without compromising function.

Change review is traceable: Record the issue, decision, affected files, evidence, owner and approval before releasing a revision.

Productivity is system design: Templates, libraries, naming, model states, automation and checklists should reduce variation across the whole team.

Working method

1. Check brief
2. Review evidence
3. Test change
4. Approve revision
5. Release package

Lab 7 — Exploded View, Storyboard and Animation

Mapping	LO6 · A6, A7 · K8
Time	70 minutes
Scenario	Explain the assembly and service order of a small engine using an exploded presentation and a concise animation.
Deliverable	An IPN presentation with trails, snapshots and storyboard plus an exported review video or image sequence.

Detailed step-by-step procedure

1. Activate the lab project, open Engine.iam from the copied engine-presentation folder and confirm all component references resolve.
2. Create a new Standard.ipn presentation and insert Engine.iam as the source model for Scene1.
3. Set a clear isometric camera and create an initial snapshot view before applying any tweaks.
4. Group components by assembly sequence and plan explode directions that do not cause visually impossible intersections.
5. Apply component tweaks in logical order, keeping related fasteners and parts on consistent axes where appropriate.
6. Enable trails selectively so they clarify movement without turning the view into clutter.
7. Create intermediate snapshot views for the major assembly stages and give them descriptive names.
8. Create a storyboard and arrange tweak actions on the timeline in the intended assembly or service order.
9. Adjust duration and start time so the viewer can follow each major stage without unnecessary delay.
10. Add a restrained camera action only if it reveals a hidden relationship; do not use motion as decoration.
11. Play the storyboard, check for collisions, confusing paths and components that move before their fasteners.
12. Edit the timeline until the static snapshots and animated sequence tell the same assembly story.
13. Create a drawing from the IPN and place an exploded view with balloons or trail visibility as required.
14. Publish a video or sequential images using a clear resolution and neutral background.
15. Save the IPN and capture the scene browser, timeline, exploded snapshot and exported output for evidence.

Acceptance criteria

- ☐ All presentation references resolve and the source IAM remains unchanged.

- ☐ Exploded paths are readable, non-colliding and consistent with a plausible assembly sequence.
- ☐ Storyboard timing and snapshots communicate the same ordered narrative.
- ☐ The exported output and drawing view are legible and suitable for stakeholder review.

Lab folder: labs/lab-07-exploded-view-storyboard-and-animation/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Lab 8 — Material, Automation and Final Design Review

Mapping	LO6 · A11 · K6, K8
Time	75 minutes
Scenario	Review an engine-block part against requirements, compare material options and encode a controlled design variant using named parameters and a simple rule.
Deliverable	A reviewed IPT variant, material decision matrix, parameter/rule evidence and signed design-review checklist.

Detailed step-by-step procedure

1. Open Engine Block Review.ipt and use Save As to create Lab08-Engine-Block-Review.ipt.
2. Read the design brief and create a review matrix covering function, envelope, interfaces, material, manufacture, safety, service and evidence.
3. Inspect model health, feature naming, sketch constraint status and update warnings; record each issue before editing.
4. Create named user parameters for two dimensions that define a controlled product variant.
5. Set realistic minimum and maximum values from the brief and test both bounds for feature failure or impossible geometry.
6. Assign the first candidate material and update mass properties; record density, mass and relevant manufacturing implications.
7. Repeat for at least two alternative materials and compare strength, stiffness, corrosion, machinability, availability, cost and mass.
8. Select a preferred material and explain the trade-off rather than choosing only the lightest or strongest option.
9. Create a simple iLogic rule or parameter check that warns when a controlled dimension is outside the approved range.
10. Run the rule at nominal, minimum and maximum values and capture the result.
11. Inspect the precision drawing and identify which views, dimensions or tolerances must be reviewed after the design change.
12. Apply one approved improvement, update the model and drawing, and record the before/after evidence.
13. Complete a DFMA review covering part count, tool access, setup count, standard features, handling and service replacement.
14. Record the issue, decision, affected files, evidence, owner and approval status in the change log.
15. Save and reopen the part, run Update and confirm the final design review has no unresolved blocker.

Acceptance criteria

- ☐ Review evidence addresses all stated requirements and clearly separates fact, assumption and recommendation.
- ☐ Material decision uses multiple performance and manufacturing criteria with traceable mass-property evidence.
- ☐ Named parameters and the rule control a valid range without failing the model.
- ☐ The approved change, affected drawing and DFMA implications are recorded before release.

Lab folder: labs/lab-08-material-automation-and-final-design-review/

Evidence: Completed native files, named screenshots, review-notes.md and the checked acceptance list.

Quick Command and File Reference

Item	Use	Release check
.ipt	Part model	Sketches and features update
.iam	Assembly	References and intended motion resolve
.ipn	Presentation	Source model and storyboard update
.idw / .dwg	Manufacturing drawing	Views, dimensions and GD&T are current
.ipj	Project definition	Workspace and libraries point to the lab

Assessment Flow and Support

1. Complete TRAQOM digital attendance.
2. Complete Assessment Digital Attendance.
3. Sit the Written Assessment, then the Practical Performance.
4. Upload the completed question papers on the LMS.
5. Sign the Assessment Summary Record.

Courseware and assessment: <https://lms-tms.tertiaryinfotech.com/>

Support: enquiry@tertiaryinfotech.com · +65 6100 0613 · www.tertiarycourses.com.sg